

Quiz 6

ECON 320 – Introduction to Mathematical Economics

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Part A — Multiple Choice (1 mark each)

1. The Envelope Theorem tells us that when evaluating the effect of a parameter on the value function:
 - (a) We must take into account the indirect effect through optimal choices.
 - (b) We ignore all partial derivatives involving the parameter.
 - (c) Only the direct partial effect matters; indirect effects cancel by FOCs.
 - (d) The derivative of the parameter is always zero.
2. In constrained optimization, the Lagrange multiplier represents:
 - (a) The marginal cost of production.
 - (b) The shadow value of relaxing the constraint by one unit.
 - (c) The total value of the objective function.
 - (d) A second-order condition checker.
3. The indirect utility function is the **dual** to:
 - (a) The expenditure function.
 - (b) The profit function.
 - (c) The production function.
 - (d) The marginal utility function.

4. If the cost function is $C(w, r, Q)$, Shephard's Lemma states:

- (a) $\frac{\partial C}{\partial Q} = MC$
- (b) $\frac{\partial C}{\partial w} = L^*$ and $\frac{\partial C}{\partial r} = K^*$
- (c) $\frac{\partial C}{\partial w} = -L^*$
- (d) $\frac{\partial C}{\partial r} = -K^*$

5. The integral $\int_0^T I(t) dt$ represents:

- (a) Total interest earned.
- (b) Total depreciation of capital.
- (c) Total accumulated investment over $[0, T]$.
- (d) Marginal change in investment.

6. A first-order linear differential equation takes the form:

- (a) $\frac{dx}{dt} = ax^2 + b$
- (b) $\frac{dx}{dt} + P(t)x = Q(t)$
- (c) $\frac{d^2x}{dt^2} + ax = 0$
- (d) $x = \int ax dt$

7. If capital evolves as $\frac{dK}{dt} = I - \delta K$, the long-run steady-state level of capital is:

- (a) $K^* = I$
- (b) $K^* = \delta I$
- (c) $K^* = \frac{1}{I}$
- (d) $K^* = \frac{I}{\delta}$

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Part B — Short Answer Problems

Question 1 — Envelope Theorem (4 marks)

Consider the constrained optimization problem:

$$\max_{x,y} U(x, y, a) \quad \text{s.t.} \quad g(x, y, a) = 0,$$

with Lagrangian:

$$\mathcal{L} = U(x, y, a) + \lambda[g(x, y, a)].$$

1. Derive the expression for the value function derivative $\frac{dV}{da}$ using the Envelope Theorem.
2. Explain (in one short paragraph) why the indirect effect through x^* and y^* drops out.
3. Provide an economic interpretation of the term $-\lambda \frac{\partial g}{\partial a}$.

Question 2 — Duality and Shephard's Lemma (4 marks)

Suppose the cost function for a firm is:

$$C(w, r, Q) = wL^*(Q) + rK^*(Q),$$

where w is the wage and r is the rental rate of capital.

1. Use Shephard's Lemma to derive the firm's conditional factor demands $L^*(Q)$ and $K^*(Q)$.
2. Briefly explain why cost-minimizing behavior implies a dual relationship between cost and production.
3. Show that marginal cost equals $\frac{\partial C}{\partial Q}$.

Question 3 — Integration and Stock–Flow Dynamics (4 marks)

Let capital evolve according to the equation:

$$\frac{dK}{dt} = I(t) - \delta K(t).$$

1. Write the integrating factor and show all steps needed to solve for $K(t)$.
2. Derive the long-run (steady-state) value of K .

3. Provide an economic interpretation of the terms:

$$K(0)e^{-\delta t}, \quad \frac{1}{\delta} \int_0^t e^{-\delta(t-s)} I(s) ds.$$

Question 4 — Present Value Calculation (4 marks)

A consumer receives income according to:

$$Y(t) = Y_0 e^{gt},$$

where $Y_0 > 0$ and g is the income growth rate. The discount rate is $r > g$.

1. Compute the present value:

$$PV = \int_0^\infty Y(t) e^{-rt} dt.$$

Show all integration steps.

2. State the condition needed for the present value to be finite and explain why.

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End of Quiz